

## I. STRATEGY OVERVIEW

Project BWC is a regime-aware quantitative portfolio tracking engine. Built to test structural alpha decay, hierarchical risk parity bisection, and topological model fusion on out-of-sample data. It employs strictly mathematical implementations (DuckDB, Polars, GraphicalLassoCV) to observe algorithmic behavioral bypass and track geometric degradation natively terminating in May 2026.

Report generated: 2026-05-25 11:28 UTC  
Initial capital: \$1,000,000 | Benchmark: SPY

## II. RISK & RETURN ANALYTICS

| Metric                  | Portfolio | Benchmark (SPY) |
|-------------------------|-----------|-----------------|
| Annualized Sharpe       | 1.30      | 0.95            |
| Annualized Sortino      | 3.44      | 1.30            |
| Calmar Ratio            | 6.19      | 1.80            |
| Maximum Drawdown        | -9.1%     | -22.1%          |
| Cornish-Fisher VaR (5%) | 2.55%     | -3.88%          |
| Conditional VaR (5%)    | -2.01%    | -4.50%          |
| Tail Ratio              | 0.95      | 0.85            |
| Annualized Return       | +67.52%   | +10.5%          |
| Annualized Volatility   | 39.69%    | 18.0%           |
| Total Return            | +7.13%    | +22.0%          |

## III. CONFIDENCE-FILTERED SIGNAL ENGINE

The signal engine operates as a monitoring layer that gates signals through a dual-threshold filter:

- Confidence minimum: 0.65
- Fused score minimum: 0.35
- Dynamic accuracy penalty: threshold increases when historical accuracy drops below 70%, making it harder for unreliable tickers to generate actionable signals.

The engine tracks realized vs. expected returns per ticker and uses exponentially-weighted rolling accuracy to ensure only high-quality signals reach the portfolio allocator.

IV. FAMA-FRENCH FACTOR ATTRIBUTION

| Factor           | Loading | Interpretation              |
|------------------|---------|-----------------------------|
| Market (MKT-RF)  | 1.36    | Significant market exposure |
| Size (SMB)       | 0.48    | Moderate small-cap exposure |
| Value (HML)      | -0.79   | Significant value exposure  |
| Momentum (UMD)   | -0.16   | Mild momentum exposure      |
| FF3 Alpha (ann.) | 0.5093  | R <sup>2</sup> = 0.43       |
| C4 Alpha (ann.)  | 0.5713  | R <sup>2</sup> = 0.43       |

V. PORTFOLIO HOLDINGS

| Ticker | Name               | Sector       | Qty  | Avg Cost | Mkt Val      | Weight |
|--------|--------------------|--------------|------|----------|--------------|--------|
| SPY    | SPDR S&P 500 ETF   | Broad Market | 173  | \$674.64 | \$128,995.72 | 12.8%  |
| AMZN   | Amazon.com         | E-commerce/C | 310  | \$221.03 | \$82,559.20  | 8.2%   |
| GLD    | SPDR Gold Shares   | Commodities  | 450  | \$436.80 | \$186,219.00 | 18.4%  |
| GOOGL  | Alphabet Inc Class | Big Tech/AI  | 210  | \$316.65 | \$80,423.70  | 7.9%   |
| PLTR   | Palantir Technolog | AI Software  | 438  | \$147.10 | \$59,953.44  | 5.9%   |
| XLU    | Utilities Select S | Utilities    | 1200 | \$45.74  | \$54,420.00  | 5.4%   |
| XLF    | Financial Select S | Financials   | 700  | \$51.25  | \$36,358.00  | 3.6%   |
| XLI    | Industrial Select  | Industrials  | 270  | \$170.75 | \$46,377.90  | 4.6%   |
| CAT    | Caterpillar Inc    | Industrials  | 100  | \$766.00 | \$87,989.00  | 8.7%   |
| JPM    | JPMorgan Chase     | Financials   | 110  | \$307.06 | \$33,701.80  | 3.3%   |
| ESGU   | iShares ESG Aware  | Broad Market | 330  | \$146.90 | \$53,697.60  | 5.3%   |
| XOM    | Exxon Mobil        | Energy       | 305  | \$154.89 | \$47,250.60  | 4.7%   |
| XLE    | Energy Select SPDR | Energy       | 60   | \$57.95  | \$3,569.40   | 0.4%   |
| IBB    | iShares Biotechnol | Healthcare   | 120  | \$167.12 | \$20,254.80  | 2.0%   |
| NVDA   | NVIDIA Corporation | AI Infrastru | 274  | \$184.04 | \$59,000.42  | 5.8%   |
| ORCL   | Oracle Corporation | Software     | 161  | \$155.76 | \$30,924.88  | 3.1%   |

## VI. WALK-FORWARD BACKTEST RESULTS

| Metric             | naive_equal_wei | topological_hrp | delta |
|--------------------|-----------------|-----------------|-------|
| total_return       | 0.3987          | 0.3406          | N/A   |
| annualized_sharpe  | 2.7902          | 3.1870          | N/A   |
| annualized_sortino | 4.1367          | 4.7970          | N/A   |
| max_drawdown       | -0.0536         | -0.0502         | N/A   |